PART IB Electromagnetism Notes

Ruize Li

Jesus College, Cambridge

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Contents

Before Everything

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1 Electrostatic Fields

The entire course will be dealing with the Maxwell equations and the Lorentz force law, which govern almost everything in electromagnetism [1]. However, we shall start at the very beginning, because there are many valuable questions we also need to deal with.

1.1 Electrostatic force

The **Coulomb Law** says that force experienced by particle 2 with respect to particle 1 was experimentally measured to be

$$\vec{F}_{12}(\vec{r}) = \frac{1}{4\pi\epsilon_0}\frac{q_1 q_2}{r_{12}^2}\hat{r}_{12}, \tag{1}$$

where $\epsilon_0 = 8.85 \times 10^{-12}$ F/m is the vacuum permittivity. The total force generated by real objects can be calculated through linear superposition of infinitesimal parts.

$$\vec{F}(\vec{r}) = \iint \frac{1}{4\pi\epsilon_0}\frac{\rho_1(\vec{r}_1)\rho_2(\vec{r}_2)}{|\vec{r}_2 - \vec{r}_1|}\hat{r}_{12}\ \mathrm{d}V_2\mathrm{d}V_1, \quad \hat{r}_{12} = \frac{\vec{r}_2 - \vec{r}_1}{|\vec{r}_2 - \vec{r}_1|}. \tag{2}$$

This is the foundation of electrostatics, and it was experimentally tested[1]. The importance of Coulomb's Law is like the fundamental theorems in Euclidean geometry. Everything follows afterwards were from mathematics and logic.

Convention of the sign of charge is defined by B. Franklin: positive charge on a glass rod after rubbing with silk tissue. Of course, particles carrying the same charge repel each other, different charge attract each other.

1.2 Potential and field

The **electric field** is defined by the force per unit charge, a vector field, which can be "measured" by an infinitesimally small test charge. The field generated by the test charge is to be excluded.

$$\vec{F}(\vec{r}) = q\vec{E}(\vec{r}) \quad \Rightarrow \quad \vec{E}(\vec{r}) = \frac{1}{4\pi\epsilon_0}\frac{q}{r^2}\hat{r}. \tag{3}$$

The **electrostatic potential** is defined by the amount of work/energy needed per **unit** electric charge to move the charge from a reference point[2] to a specific point in an electric field.

$$\mathrm{d}V = -\mathrm{d}\vec{r}\cdot\vec{E}(\vec{r}), \quad V_{BA} = -\int_A^B \vec{E}(\vec{r})\cdot\mathrm{d}\vec{r}, \quad \Leftrightarrow \quad \vec{E} = -\nabla V(\vec{r}). \tag{4}$$

Because the "zero point" is arbitrarily defined, potential is not a physical observable, and only potential difference matters.

Once the point (**gauge**) is defined, potential at each point in space can be calculated using line integrals, and behaves as a scalar field. Since the field is **conservative**[3], the line integral does not depend on the path chosen. Hence, the potential is **a property of the space** in which the test charge moves, rather than being a property of the way in which it moves.

$\vec{E} = -\nabla V(\vec{r})$ points in the direction in which the potential decreases most rapidly with position, normal to the contour passing through that point.

1.3 Integral theorems

From Stokes's theorem, we know that electrostatic field satisfies

$$\oint_{\partial S}\vec{E}\cdot\mathrm{d}\vec{l} = \int_S(\nabla\times\vec{E})\cdot\mathrm{d}\vec{S}. \tag{5}$$

And by Gauss's law (divergence theorem),

$$\oint_{\partial V}\vec{E}\cdot\mathrm{d}\vec{S} = \int_V \nabla\cdot\vec{E}\ \mathrm{d}V = \int_V \frac{\rho(\vec{x})}{\epsilon_0}\ \mathrm{d}V. \tag{6}$$

The result implies that electric fields have net sources (charge, or monopoles). Writing it in a simpler way, we define $\vec{D} = \epsilon_0\vec{E}$ as the electric displacement[4] and write

$$\nabla\cdot\vec{D} = \rho(\vec{r}), \quad \Leftrightarrow \quad \oint_{\partial V}\vec{D}\cdot\mathrm{d}\vec{S} = \int_V \rho(\vec{r})\ \mathrm{d}V. \tag{7}$$

These equations won't make things harder (only) if there exists some prefered symmetry.

[1] The tests on Coulomb's law remains the most precise experiment ever.

[2] Normally chosen at infinity.

[3] Only when energy is conserved, and it leads to $\frac{\partial E_i}{\partial x_j} = \frac{\partial E_j}{\partial x_i}$

[4] This does not make any sense without polarisation, of course.

1.4 Multipole expansion

Expanding the $1/r$ in the electrostatic potential

$$\phi(\vec{x}) = \int_V \frac{1}{4\pi\epsilon_0} \frac{\rho(\vec{x})}{|\vec{r}|} \, \mathrm{d}V, \tag{8}$$

where we have assumed that charges gather near the origin. It gives

$$\phi(\vec{x}) = \frac{1}{4\pi\epsilon_0} \int_V \rho(\vec{x}) \left[\frac{1}{R} - \vec{x} \cdot \nabla \frac{1}{R} + \frac{1}{2} \sum_{i,j=1}^{3} x_i x_j \frac{\partial^2}{\partial x_i \partial x_j} \frac{1}{R} + \cdots \right] \, \mathrm{d}V. \tag{9}$$

The results would be a Laurent series, by far we only care about the first two terms, which are the electrostatic monopoles and dipoles. For any general charge distribution, every term in the infinite series may exist. Remember that the charge distribution here is close to the origin, the monopole term and dipole term are

$$Q = \int \rho(\vec{x}) \, \mathrm{d}V, \quad \vec{p} = \int \vec{x} \rho(\vec{x}) \, \mathrm{d}V. \tag{10}$$

Higher-order terms are tensors which will not be considered here.

There is a slight problem that we haven't prove that potentials add. However, think about its definition

$$V = -\int \vec{E} \cdot \mathrm{d}\vec{x},$$

the direct superposition shouldn't be surprising. Therefore, with the limit $a \to 0$, the potential of a dipole (centred at the origin, oriented in the $\hat{r}$ direction) should be

$$V_{\text{dipole}} = \frac{1}{4\pi\epsilon_0} \frac{p \cos\theta}{r^2}, \quad p = qa. \tag{11}$$

And therefore the electric field produced is

$$\vec{E} = -\nabla V = \frac{p}{4\pi\epsilon_0} \frac{2\cos\theta \hat{e}_r + \sin\theta \hat{e}_\theta}{r^3}. \tag{12}$$

In a locally uniform electric field E, the potential energy is

$$U = -\vec{p} \cdot \vec{E}, \tag{13}$$

which gives a $\cos\theta$ expression and implies that the potential energy is minimum when the dipole is aligned with the field. The force applied on the dipole is

$$\vec{F} = (\vec{p} \cdot \nabla)\vec{E}, \quad \Leftrightarrow \quad F_j = p_i \frac{\partial E_j}{\partial x_i}. \tag{14}$$

If the dipole moment is constant, we can take it into the derivative and write $\vec{F} = \nabla[\vec{p} \cdot \vec{E}]$, which satisfies $\vec{F} = -\nabla U$. However, this should not be confused with $\vec{F} = -\nabla V$ that we wrote before, and cannot be regarded as the definition of electric potential energy. The torque (couple) acting on the dipole is

$$\vec{M} = \vec{p} \times \vec{E}. \tag{15}$$

Of course dipole would explain why charged spoon can attract water flow, but quantitative calculation would require knowledge about polarisation.

1.5 Electric Flux

Electric flux defines the density of field lines (introduced by Faraday and explained mathematically by Maxwell [2]), and is defined by

$$\Phi_E = \int_S \vec{E}(\vec{r}) \cdot \mathrm{d}\vec{S}. \tag{16}$$

Divergence measures the net outflow from a point per unit volume.[5]

[5] And the curl tells us the line integral around a small area at a point.

1.6 Charge Conservation

Charge conservation is a natural law and is one of the things that no experiment had yet challenged. Applying integral over a certain volume

$$\int_{\partial V} \vec{J} \cdot \mathrm{d}\vec{S} = \int_V \nabla \cdot \vec{J} \, \mathrm{d}V = 0, \tag{17}$$

we have

$$\frac{\partial \rho}{\partial t} + \nabla \cdot \vec{J} = 0. \tag{18}$$

The **continuity equation** is used in almost all topics (including QFT) because it is a direct result from **normalisation**.

1.7 Laplace's and Poisson's Equations

We want to know how is the potential at one point related to the potential at a neighbouring point. Which is simply the differential equation for electric potential. For a general problem with charge distribution $\rho(\vec{r})$, solutions may be given by the Poisson's equation,

$$\nabla \cdot \vec{E}(\vec{r}) = -\nabla^2 V(\vec{r}) = \frac{\rho(\vec{r})}{\epsilon_0}. \tag{19}$$

If $\rho(\vec{r}) = 0$, it turns into the Laplace equation

$$\nabla^2 V = 0. \tag{20}$$

1.8 Green's Function

Green's function is a mathematical tool to solve Poisson's equation[6],

$$V(\vec{r}) = \frac{1}{\epsilon_0} \int G(\vec{r}, \vec{r}')\rho(\vec{r}') \, \mathrm{d}^3\vec{r}', \tag{21}$$

where $G(\vec{r}, \vec{r}')$ is called the Green's function of the differential equation. This equation shows that the potential at r is equal to the sum of the potentials that would exist at r if the charge density distribution were divided into infinitesimally small volumes that contain the appropriate infinitesimal amounts of charge.

The Green's function is the solution of the equation

$$\nabla^2 G(\vec{r}, \vec{r}') = -\delta(\vec{r} - \vec{r}') \tag{22}$$

that satisfies 'homogeneous' boundary conditions ($V(\infty) = V'(\infty) = 0$). Once the Green's function is known for the system of interest, defined in terms of the boundary conditions, the potential function can be found for any distribution of charge within the system.

1.9 Boundary Conditions and Uniqueness

- Dirichlet: the quantity of interest is specified over the boundary.
- Neumann: the normal derivative of the quantity of interest is specified over the boundary.
- Cauchy: Mixed.

Uniqueness: the problem is well-posed if either Dirichlet or Neumann boundary conditions are specified over the whole closed surface[7].

However, solutions from Neumann conditions are underdetermined within an additive constant since they were built from derivatives.

[6] Refer to mathematics notes.
[7] Infinity counts.

Proof: Suppose we have two solution to the same Poisson's equation,

$$\nabla^2 V = -\frac{\rho}{\epsilon_0};$$
$$\nabla^2 U = -\frac{\rho}{\epsilon_0}.$$

$\Phi = V - U$ is therefore a solution of Laplace's equation,

$$\nabla^2 \Phi = \nabla^2 V - \nabla^2 U = 0.$$

We can now use the vector differential identity

$$\nabla \cdot (\Phi \nabla \Phi) = |\nabla \Phi|^2 + \Phi \nabla^2 \Phi,$$

and integrate over the total volume

$$\int \mathrm{d}V \ \nabla \cdot (\Phi \nabla \Phi) = \int_S \mathrm{d}\vec{S} \cdot (\Phi \nabla \Phi).$$

If the Dirichlet boundary conditions are satisfied by both solutions, $\Phi = 0$ on the boundary, and

$$\int \mathrm{d}V |\nabla \Phi|^2 = 0,$$

which gives $|\nabla \Phi| = 0$. Since Φ is zero on the boundary, it must be everywhere zero. Similarly, if Neumann conditions are satisfied, the normal component of $\nabla \Phi$ would be zero on the boundary and the result would be the same, apart from a constant k.

For a conducting sphere in a uniform field, uniqueness tells us that the extra field from charge distribution on its surface is the same as the field generated by a dipole,

$$p = 4\pi\epsilon_0 r^2, \tag{23}$$

where r is the radius of the sphere. Now we shall introduce the concept of polarisability, which is defined by

$$\vec{p} = \alpha \vec{E}. \tag{24}$$

Polarisability is, of course, a tensor generally. But in conductors, by symmetry, it appears as a scalar.

This example may convince us to believe the method of images. Note that it's all about symmetry and simulation of charge distributions.

Method of images is particularly useful when solving the charge distribution, capacitance, and the geometry of field.

1.10 Electrostatic Energy

Energy = charge × potential. We build our equation for potential energy by[8]

$$U = \sum_{j=1}^{N} \sum_{i<j} \frac{q_i q_j}{4\pi\epsilon_0 r_{ij}} = \frac{1}{2} \sum_{j=1}^{N} \sum_{i \neq j} \frac{q_i q_j}{4\pi\epsilon_0 r_{ij}} = \frac{1}{2} \sum_{j=1}^{N} q_j \sum_{i \neq j} \frac{q_i}{4\pi\epsilon_0 r_{ij}} = \frac{1}{2} \sum_{j=1}^{N} q_j V_j. \tag{25}$$

If the charge distribution is continuous, it changes to

$$U = \frac{1}{2} \int \rho(\vec{r}) V(\vec{r}) \ \mathrm{d}^3\vec{r}, \tag{26}$$

integrated over the whole space.

[8]The charge cannot feel its own force!

Note that, this is a potential energy caused by interactions, which does not include the energy of electric field in vacuum. Derivation of the energy density of the field is nothing trivial. Normally it is explained by calculating the energy density inside a parallel plane capacitor. However, note that the work required to assemble a charge distribution is [3]

$$W_e = \frac{\epsilon_0}{2} \int E^2 \, \mathrm{d}^3 r, \tag{27}$$

where

$$U_E(\vec{r}) = \frac{1}{2}\epsilon_0 \vec{E}^2 \tag{28}$$

is the energy density of the electric field[9]. This can be derived using Gauss's law and the divergence theorem,

$$\begin{aligned} U &= \frac{1}{2} \int \mathrm{d}^3\vec{r} \; \rho(\vec{r}) V(\vec{r}) \\ &= \frac{1}{2}\epsilon_0 \int \mathrm{d}^3\vec{r} \; [\nabla \cdot \vec{E}(\vec{r})] V(\vec{r}) \\ &= \frac{1}{2}\epsilon_0 \int \mathrm{d}^3\vec{r} \; [\nabla \cdot [V(\vec{r})\vec{E}(\vec{r})] - \vec{E}(\vec{r}) \cdot \nabla V(\vec{r})] \\ &= -\frac{1}{2}\epsilon_0 \int \mathrm{d}^3\vec{r} \; \vec{E}(\vec{r}) \cdot \nabla V(\vec{r}) \\ &= \frac{1}{2}\epsilon_0 \int \mathrm{d}^3\vec{r} \; |\vec{E}(\vec{r})|^2. \end{aligned} \tag{29}$$

This may be confusing at first glance, because here W_e is always positive, but the energy defined by Eq. 26 can be negative. That's because we're considering different physical processes here. Eq. 26 does not take into account the work necessary to make the point charges in the first place; we started with point charges and simple found the work required to bring them together [4].

We shall not repeat force and virtual work because that should be covered in the Classical Dynamics course.

1.11 Capacitance

The capacitance of an isolated object was defined by

$$C = \frac{Q}{V}, \tag{30}$$

where Q is the charge on that object and V is the voltage of it from our reference point. If the capacitor is made up from two objects, then V is the potential difference between them.

To calculate the energy stored in a capacitor, we divide the capacitor into parts, such that,

$$Q = \sum_{i=1}^{N} \mathrm{d}Q_i,$$

and use Eq. 25. That gives,

$$U_N = \frac{1}{2} \sum_{i=1}^{N} \mathrm{d}Q_i \frac{Q - \mathrm{d}Q_i}{C} = \frac{1}{2} N \frac{Q \; \mathrm{d}Q}{C} - \frac{1}{2} N \frac{(\mathrm{d}Q)^2}{C}.$$

Notice that $N \; \mathrm{d}Q = Q$, and ignore the second-order term, we get

$$U = \frac{1}{2}\frac{Q^2}{C} = \frac{1}{2} C V^2 = \frac{1}{2} Q V. \tag{31}$$

Another energy that can be calculated is the energy stored in the **field** of the capacitor. This can only be done after we have worked out the field itself, because we need to integrate the energy density

$$\epsilon = \frac{1}{2}\vec{D} \cdot \vec{E}$$

over the whole space. We have shown in the last subsection that the two energies must have the same value.

[9] Should be $\frac{1}{2}\vec{D} \cdot \vec{E}$ with polarisation.

1.12 Force and virtual work

Virtual work is a useful concept when dealing with perturbations at equilibrium, and is widely used in engineering cases. Mathematically, virtual work is

$$F \cdot \mathrm{d}x = \frac{\partial U_s}{\partial x} \cdot \mathrm{d}x + \frac{\partial U_d}{\partial x} \cdot \mathrm{d}x, \tag{32}$$

where U_s is the energy stored in the electrostatic field, and U_d is the energy dissipated in some way (such as current through a resistor).

2 Electrostatic Fields in Dielectric Materials

2.1 Polarisation

Faraday found that when an insulator is placed between the plates of a capacitor held at constant potential difference, the charge of the plates increases. This effect is called polarisation. The increase is by a factor of ϵ_r, relative permittivity, a dimensionless constant which describes the dielectric property of the material.

In insulators, charge cannot flow freely in the material, so all charge is **bound** instead of **free**. An applied electric field can separate positive and negative bound charge, creating dipole moments, resulting in a charge distribution with total net charge zero. As one can imagine, the dipoles inside the insulator would cancel each other, therefore macroscopically, charge only appears on the external surfaces.

In reality, the dipoles are caused by different reasons, depends on the molecular structure and alignment of molecules. For instance, water molecules has net dipole moment thus they just need to realign towards the opposite of field, gases such as HCl behave similarly. In contrast, NaCl crystal (structure discussed in PART IA Materials Science) creates net dipole by shifting the position of Na and Cl molecules slightly in opposite directions. In general, materials are **polarised** by a combination of these two effects, but one of them may be neglect-able.

One thing that may cause a slight problem is that, all materials have tolerance to electric fields. High electric fields, even just a local one, can cause penetration, and the material loses its properties as an insulator for a short time. This is normally to be avoided in engineering cases.

It is usual to express the net dipole moment per unit volume as $\vec{P} = n\vec{p}$, often referred to as **polarisation**, because it measures how much the material is polarised. By its definition, it is clear that once $\vec{P}$ is known, the polarisation charge density at the surface with normal $\hat{n}$ is $\sigma = \vec{P} \cdot \hat{n}$.

Note that a parallel plate capacitor can be modelled as a sheet of dipoles, which gives us the intuition to compare $\vec{P}$ to electric fields. By Gauss's law,

$$\nabla \cdot \vec{P} = -\rho_{\mathrm{p}}, \tag{33}$$

where ρ_{p} is the density of polarised charge, integrated over the whole space, it should give zero. Now we shall see the convenience that $\vec{D}$ has brought us,

$$\begin{aligned}
\oint \mathrm{d}\vec{S} \cdot \vec{E} &= \frac{1}{\epsilon_0} \int \mathrm{d}V \ (\rho_{\mathrm{f}} + \rho_{\mathrm{p}}), \\
\int \mathrm{d}V \ \nabla \cdot \vec{E} &= \frac{1}{\epsilon_0} \int \mathrm{d}V \ (\rho_{\mathrm{f}} - \nabla \cdot \vec{P}), \\
\int \mathrm{d}V \ \nabla \cdot [\epsilon_0 \vec{E} + \vec{P}] &= \int \mathrm{d}V \ \rho_{\mathrm{f}}, \\
\nabla \cdot [\epsilon_0 \vec{E} + \vec{P}] &= \nabla \cdot \vec{D} = \rho_{\mathrm{f}}.
\end{aligned} \tag{34}$$

Therefore, polarisation relates to bound charge, electric field relates to total charge, and the displacement field relates to free charge.

Since ϵ_r is material-specific, as long as the proportional relation holds (in experiments[10]), we are free to define $\vec{P} = \epsilon_0(\epsilon_r - 1)\vec{E}$, which means

$$\vec{D} = \epsilon_0 \epsilon_r \vec{E}. \tag{35}$$

Sometimes people write $\epsilon_r - 1$ as susceptibility χ, but I suggest that we keep the ϵ_r convention.

The vector field $\vec{D}$ is called the **electric displacement** or the **electric flux density**.

Note that, we have modified Gauss's Law, but we haven't checked if Stokes's law has changed due to polarisation. By linear superposition, it's clear that

$$\nabla \times \vec{D} = \nabla \times \vec{P}.$$

The effect can be seen, for example, in the field created by an electret.

Note that, we have only considered dipoles in the materials, which is the lowest-order term from a series of possibilities. In fact, the gradient of the field induces a quadrupole, the second derivative induces an octupole, etc. In most circumstances, the first term dominant, which keeps our assumptions safe in numerical calculations.

2.2 Boundary Conditions Again

At the boundary between two materials of dielectric constants ϵ_1 and ϵ_2, applying Gauss's law, it is easy to show that

$$\oint \vec{D} \cdot \mathrm{d}\vec{S} = \int \rho_\mathrm{f} \ \mathrm{d}V = 0, \quad \Rightarrow \quad \vec{D}_{2,\perp} = \vec{D}_{1,\perp}. \tag{36}$$

And, by Stokes's theorem,

$$\oint \vec{E} \cdot \mathrm{d}\vec{l} = 0, \quad \Rightarrow \quad \vec{E}_{2,\parallel} = \vec{E}_{1,\parallel}. \tag{37}$$

Conclusion:

- The component of $\vec{D}$ perpendicular to the surface is continuous;
- The component of $\vec{E}$ parallel to the surface is continuous.

2.3 Example: Dielectric Sphere in Uniform External Field

Suppose a dielectric sphere is placed in a uniform external field $E_0\hat{z}$, and the field inside the sphere is $E_\mathrm{in}\hat{z}$. By this we are saying that the sphere is uniformly polarised. We also assume that the total field outside is the superposition of the uniform field E_0 and a dipole field $V_p \propto \frac{\kappa \cos\theta}{r^2}$, therefore,

$$\begin{aligned} V_\mathrm{in} &= -E_\mathrm{in} r \cos\theta; \\ V_\mathrm{out} &= -E_0 r \cos\theta + \frac{\kappa \cos\theta}{r^2}. \end{aligned} \tag{38}$$

Applying the boundary conditions, we require that $E_\parallel$ is continuous at $r = a$,

$$-\frac{1}{r}\frac{\partial V}{\partial \theta}\bigg|_{r=a} = -E_\mathrm{in}\sin\theta = -E_0 \sin\theta + \frac{\kappa \sin\theta}{a^3}.$$

Similarly, we require that $D_\perp$ is continuous at $r = a$, therefore,

$$-\epsilon\epsilon_0 \frac{\partial V}{\partial r}\bigg|_{r=a} = \epsilon\epsilon_0 E_\mathrm{in}\cos\theta = \epsilon_0 E_0 \cos\theta + \epsilon_0 \frac{2\kappa\cos\theta}{a^3}.$$

Combining the two equations, we have

$$\kappa = \frac{\epsilon - 1}{\epsilon + 2} a^3 E_0; \quad E_\mathrm{in} = E_0 - \frac{\epsilon - 1}{\epsilon + 2} E_0. \tag{39}$$

[10]Normally it is, but there are non-linear cases, most widely present in high-field experiments and optical phenomena.

3 Magnetostatic Fields

It is an experimental fact that electrical current gives rise to magnetic fields, and magnetostatics describes how magnetic fields generated by stationary field source behave in vacuum and specific materials.

3.1 Magnetostatic Force

We define the current element by a filament of wire of infinitesimal length carrying current I, and we take the current density's direction $\hat{j}$ as it's unit vector:

$$\text{current element} = I \ \mathrm{d}l\hat{j} = I \ \mathrm{d}\vec{l}. \tag{40}$$

The magnetic field $\vec{B}$ at position $\vec{r}$ is defined through the force $\vec{F}$ on an imaginary current element,

$$\mathrm{d}\vec{F} = I \ \mathrm{d}\vec{l} \times \vec{B}. \tag{41}$$

This is consistent with the Lorentz force on a moving charge,

$$I \ \mathrm{d}\vec{l} = \frac{\mathrm{d}q}{\mathrm{d}t}\vec{v} \ \mathrm{d}t = \mathrm{d}q \ \vec{v} \quad \mathrm{d}\vec{F} = \mathrm{d}q \ \vec{v} \times \vec{B}. \tag{42}$$

The unit of magnetic field is Tesla (T).

Note that, unlike electric field, current element has an orientation which takes part in the representation of force.

Conversely, the magnetic field produced by a current element $\mathrm{d}\vec{l}$ is given by the Biot-Savart law. Similar to Coulomb's law, magnetic field is proportional to $1/r^2$:

$$\mathrm{d}\vec{B} = \frac{\mu_0}{4\pi}\frac{\mathrm{d}\vec{l} \times \hat{r}}{r^2}, \tag{43}$$

where μ_0 is a constant called the permeability of free space, its value depends on our chosen unit system. In SI units, $\mu_0 = 4\pi \times 10^{-7}\ \mathrm{Hm}^{-1}$.

The force between two current element is therefore

$$\mathrm{d}\vec{F} = \frac{\mu_0}{4\pi}\frac{\mathrm{d}\vec{l_2} \times (\mathrm{d}\vec{l_1} \times \hat{r}_{12})}{r_{12}^2}, \tag{44}$$

note that $\mathrm{d}\vec{l_1}$ and $\mathrm{d}\vec{l_2}$ are not associative.

Unlike the Gauss's law for electric fields which rises from the fact that electric fields are conservative, Gauss's law for magnetic fields rises from the fact that no magnetic monopole in any form was observed. Therefore, we must have

$$\nabla \cdot \vec{B}(\vec{r}) = 0 \quad \Leftrightarrow \quad \oint_S \mathrm{d}\vec{S} \cdot \vec{B}(\vec{r}) = 0. \tag{45}$$

The fact that magnetic monopole does not exist has a considerable importance, it affects almost any theory of the universe, even the Lagrangian density of the standard model.

Also, from the form of Biot-Savart law, it is easy to show that Ampere's law holds[11]:

$$\oint_{\partial S} \mathrm{d}\vec{l} \cdot \vec{H} = \int_S \mathrm{d}\vec{S} \times \vec{J} \quad \Leftrightarrow \quad \nabla \times \vec{H} = \vec{J}. \tag{46}$$

The reason why we wrote $\vec{H} = B/\mu_0$ instead of $\vec{B}$ is to be consistent with the Ampere's law in magnetic materials.

[11] Historically, Ampere's law is experimentally found.

3.2 Magnetic Scalar Potential

Historically, people tried to explain magnetism in natural magnets by using the "molecular current" approach. It assumes that each molecule produces a net current loop around it, which appears like a magnetic dipole. This makes sense classically, but from quantum mechanics we know that electrons can't actually go around molecules. Though tons of experiments were performed and tons of paper written, how magnetism was generated in materials remains a problem to be solved.

We define the magnetic dipole moment by

$$\mathrm{d}\vec{m} = I\mathrm{d}\vec{S}, \tag{47}$$

where $\mathrm{d}\vec{S}$ is the vector area perpendicular to the infinitesimal current loop. The couple on a magnetic dipole moment is therefore

$$\mathrm{d}\vec{G} = \mathrm{d}\vec{m} \times \vec{B}, \tag{48}$$

and the force

$$\mathrm{d}\vec{F} = (\mathrm{d}\vec{m} \cdot \nabla)\vec{B}, \tag{49}$$

which are the same as electric dipoles. The potential energy of that dipole is simply $U = -\vec{m} \cdot \vec{B}$, which allows us to define the magnetic scalar potential.

We define the magnetic scalar potential to be

$$\mathrm{d}\phi_m(\vec{r}) = \frac{\mathrm{d}\vec{m} \cdot \vec{r}}{4\pi r^3}, \tag{50}$$

which implies that, the magnetic scalar potential for a macroscopic loop is

$$\phi_m = \frac{I\Omega}{4\pi}. \tag{51}$$

3.3 Magnetic Vector Potential

From Gauss's and Ampere's laws, magnetic field lines are always continuous and forms a loop. Therefore, magnetic field is not conservative, which means that it cannot be described as the gradient of a (differentiable, one-one) scalar field.[12] However, Biot-Savart law allows us to express magnetic field using a vector potential $\vec{A}$,

$$\vec{B}(\vec{r}) = \nabla \times \vec{A}(\vec{r}) \quad \Leftrightarrow \quad \vec{A}(\vec{r}) = \frac{\mu_0}{4\pi} \int \frac{\vec{J}(\vec{r}')}{|\vec{r} - \vec{r}'|} \, \mathrm{d}^3\vec{r}'. \tag{52}$$

However, the choice of $\vec{A}$ is not unique, we can add any vector field $\vec{K}(\vec{r})$ that satisfies $\nabla \times \vec{K}(\vec{r}) = 0$ to this without changing the resultant $\vec{B}$. Therefore we must choose a gauge for the vector potential. Usually the choice is

$$\nabla \cdot \vec{A}(\vec{r}) = 0, \tag{53}$$

which is called the Coulomb gauge. In electrodynamics, particularly when there are electromagnetic waves presenting, we shall choose the Lorenz gauge, such that

$$\nabla \cdot \vec{A} + \frac{1}{c^2}\frac{\partial \phi}{\partial t} = 0. \tag{54}$$

The Poisson equation of magnetic vector potential, follows directly from $\nabla \times \vec{B} = \mu_0 \vec{J}$, $\vec{J} = \sigma \vec{E}$, is (in Coulumb gauge)

$$\nabla^2 \vec{A} = -\mu_0 \vec{J}. \tag{55}$$

[12] Unless the system is built from dipole moments only, where we have unconsciously introduced two monopoles and "closed the loop" by placing them together.

4 Magnetostatic Fields in Magnetic Materials

We've covered the basis of this in Part IA materials and did a few experiments on hysteresis and phase transformations. Therefore, we should be comfortable with the concepts of diamagnetic, paramagnetic and ferromagnetic materials.

Diamagnetic means that the closed current loops oppose any applied external magnetic field. Paramagnetic means the permanent dipole moments of the material align themselves with the external field. Ferromagnetic materials reacts similarly, but the effect is considerable enhanced.

The different properties between the three kinds of materials are from the precise nature of the magnetic dipole moments of the atoms and molecules that make up the materials. Two main contributions to the magnetic dipole moment of an atom or molecule are orbital motions of electrons and the intrinsic dipole moments of electrons from their spin, and the overall effect should be deterimined by the sum of these two factors. We shall not discuss them in detail in a proper physics course.

4.1 Magnetisation

Similar to polarisation, we define the magnetisation as the magnetic dipole moment per unit volume,

$$\vec{M} = \frac{\vec{m}}{\mathrm{d}x\mathrm{d}y\mathrm{d}z}. \tag{56}$$

Therefore the total magnetic dipole moment of an object is

$$\vec{m}_{\mathrm{tot}} = \int_V \mathrm{d}^3\vec{r}\ \vec{M}.$$

To be distinguished from the movement of free charge, the moves of current loops can be measured using $\vec{J}_m$, the magnetisation current density, which is determined by

$$\vec{J}_m = \nabla \times \vec{M}. \tag{57}$$

Of course, $\vec{J}_m$ appears generally on the surface of materials, because the loops inside sums to zero.

The total magnetic field strength is therefore naturally

$$\mu_0\vec{H} = \vec{B} - \mu_0\vec{M}, \quad B = \mu_0\mu H, \tag{58}$$

where μ is the relative permeability.

Note that, most ferromagnetic materials are non-linear for relatively low field strengths and highly hysteretic. Some may be permanently magnetised even without external fields. They are characterised by their constant dipole moment per unit volume.

4.2 Boundary Conditions for Magnetic Fields

Similar to electric fields, we apply Gauss's law and Ampere's law to a small pillbox on the surface of two median. It is easy to find that

$$\begin{aligned} \oint \mathrm{d}\vec{S}\cdot\vec{B} = 0 \quad &\Rightarrow \quad \vec{B}_\perp \text{ continuous;} \\ \oint \mathrm{d}\vec{l}\cdot\vec{H} = 0 \quad &\Rightarrow \quad \vec{H}_\parallel \text{ continuous.} \end{aligned} \tag{59}$$

5 Electromagnetic Induction

We want to find out what happens if EM fields change with time. But we have to assume that this change is subject to field only, and relatively slow compared to the speed of light.[13] The cases for fast changes, such as electron moving at high speed, will be discussed later in PART II.

[13]The term "quasi-static" may be appropriate.

5.1 Faraday's Law

Faraday found experimentally that electric potential was induced when the magnetic flux around a wire loop changes. In the early 1800s this experiment was done by inserting (withdrawing) a rod magnet to (from) a solenoid. Mathematically, this can be written as

$$\frac{\mathrm{d}\Phi_\mathrm{B}}{\mathrm{d}t} = -\epsilon, \tag{60}$$

where the minus sign implies Lenz's law:

> *The magnetic field generated by the induced electric field must oppose the original magnetic field that induces the electric field.*

This may cause confusion at first glance, but will soon be clear after a few examples.

In 1832, Henry discovered self-inductance: A charge in the current flowing in a coil of wire induces an e.m.f. across its own terminals. Self-inductance is therefore defined by

$$L = \frac{\Phi_\mathrm{B}}{I}. \tag{61}$$

And the voltage caused by self-inductance is

$$V_\mathrm{gap} = L\frac{\mathrm{d}I}{\mathrm{d}t}. \tag{62}$$

This is true for any circuit, but the amount of self-inductance in most circuits are negligible.

Using integrals and differentiation, Fadaday's law can be expressed by

$$\oint \mathrm{d}\vec{l} \cdot \vec{E} = -\int \mathrm{d}\vec{S} \cdot \frac{\partial \vec{B}}{\partial t}. \tag{63}$$

Actually the RHS should be $\int \mathrm{d}\left(\frac{\partial \Phi_\mathrm{B}}{\partial t}\right)$, but if we are integrating across a non-varying surface, the vector area can be withdrawn from the brackets. The second Maxwell's equation becomes

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} \quad \Leftrightarrow \quad \oint \mathrm{d}\vec{l} \times \vec{E} = \int \mathrm{d}\vec{S} \cdot \frac{\partial \vec{B}}{\partial t}. \tag{64}$$

An alternative view of Faraday's law is to consider the forces on the free charges in a loop when the loop is moving through the magnetic field. But this depends on the quasi-static assumption, and we will not discuss this into details.

5.2 Inductor

As we already knew, the energy stored in an inductor with inductance L is

$$U = \int \mathrm{d}^3\vec{r}\, \frac{1}{2}\vec{B} \cdot \vec{H} = \frac{1}{2}\Phi I = \frac{1}{2}LI^2, \tag{65}$$

But this is only self-inductance, how about mutual inductance between two loops? The total energy is expressed by

$$U = \frac{1}{2}L_1 I_1^2 + \frac{1}{2}L_2 I_2^2 + M_{12} I_1 I_2, \tag{66}$$

where $M_{12} = M_{21}$ is the mutual inductance between the inductors. Note that it is defined by energy stored in it.

Naturally, $M^2 = L_1 L_2$ should hold, but this only happens as we have zero magnetic flux leakage, which is impossible in engineering. Therefore we introduce a coupling coefficient k to account for the remaining proportion of flux, $0 \leq k \leq 1$, and write

$$M^2 = k^2 L_1 L_2. \tag{67}$$

5.3 The ideal transformer

The ideal transformer consists of two coils coupled by a core of ferromagnetic material, with no magnetic leakage ($k = 1$). We also assume that the wires are perfectly conductive, and the core exhibits no hysteresis.

The linked fluxes Φ_1 and Φ_2 becomes

$$\Phi_1 = N_1\Phi, \quad \Phi_2 = N_2\Phi \tag{68}$$

because they share the same magnetic flux. Therefore by Faraday's law,

$$\frac{V_1}{V_2} = \frac{N_1}{N_2}. \tag{69}$$

The self-inductance of the individual coils are

$$L_{1,2} = \mu\mu_0 \left(\frac{N_{1,2}}{l_{1,2}}\right)^2 A l_{1,2} \tag{70}$$

and therefore

$$\frac{L_1}{L_2} = \frac{N_1^2/l_1}{N_2^2/l_2}. \tag{71}$$

The choices of N_1 and N_2 are often related to practical engineering considerations, rather than pure mathematics in those equations.

Of course, by energy conservation the power VI must be preserved by the ideal transformer.

5.4 Transformer circuits

In a real transformer circuit we should include the mutual inductance. Assume a sign convention: when the primary and secondary currents both flow into the top of the transformer, the resultant fluxes add. We want to calculate the current I_1 in the primary circuit when the secondary impedance Z_2 and primary voltage V_1 are known. The total flux Φ_1 is

$$\Phi_1 = L_1 I_1 - M I_2, \tag{72}$$

which leads to V_1 and V_2. Since the primary and secondary currents must be in-phase, we can assume a angular frequency ω, and write

$$\mathrm{i}\omega M I_1 = (Z_2 + \mathrm{i}\omega L_2) I_2. \tag{73}$$

Using $M = \sqrt{L_1 L_2}$, we have

$$Z_1 = \frac{\mathrm{i}\omega L_1 Z_2 (N_1/N_2)^2}{\mathrm{i}\omega L_1 + Z_2 (N_1/N_2)^2}. \tag{74}$$

For a typical electric device, $\omega L_1 \gg Z_2(N_1/N_2)^2$, therefore we have

$$Z_1 \simeq Z_2 (N_1/N_2)^2 \tag{75}$$

which is the load impedance ratio across the transformer.

5.5 Energy flow in resonant circuits

Consider a LCR circuit, in series. It is clear from the differential equation

$$V = R\frac{\mathrm{d}q}{\mathrm{d}t} + \frac{q}{C} + L\frac{\mathrm{d}^2 q}{\mathrm{d}t^2}$$

that:

- At times when energy is flowing into the magnetic field of the inductor, it is flowing out of the electric field of the capacitor, and vice versa.
- Energy always flows into the resistor.

The power flow can be calculated by considering the voltage and current relations.

The resonance frequency is $\omega_0 = 1/\sqrt{LC}$. If we are below the resonant frequency, the overall circuit looks capacitive because almost all energy was stored in the capacitor. Similarly, at high frequencies, the overall circuit looks inductive. Precisely at the resonant frequency, the energy flow stops **at all times**, and the circuit looks like a single resistor of value R.

All physical arrangements have equivalent circuits that can be analysed easily. For example, we have used LCR circuits to account for the wires in our practical sessions.

5.6 Magnetic energy

For a system with a lot of "current loops", the magnetic energy stored is

$$U = \sum_{i \in \text{loops}} \frac{1}{2} \Phi_i I_i. \tag{76}$$

In this equation, both *self-energies* and *interaction energies* are included, because magnetic flux accounts for all sources. This is the energy needed to assemble the system.

If we re-write the flux using Stokes' theorem,

$$\Phi = \int \mathrm{d}\vec{S} \cdot \vec{B} = \oint \mathrm{d}\vec{l} \cdot \vec{A}, \tag{77}$$

we have

$$U = \frac{1}{2} = \sum_{i \in \text{loops}} \frac{1}{2} \int_i \mathrm{d}^3\vec{r}\ \vec{J} \cdot \vec{A}. \tag{78}$$

If we want to go further, writing $\vec{J}$ as $\nabla \times \vec{H}$, and using the triple product identity, we will get (combining volume integrals over all loops)

$$\begin{aligned} U &= \frac{1}{2} \int \mathrm{d}^3\vec{r}\ \vec{A} \cdot (\nabla \times \vec{H}) \\ &= \frac{1}{2} \int \mathrm{d}^3\vec{r}\ \vec{H} \cdot (\nabla \times A) - \frac{1}{2} \int \mathrm{d}^3\vec{r}\ \nabla \cdot (\vec{A} \times \vec{H}) \\ &= \frac{1}{2} \int \mathrm{d}^3\vec{r}\ \vec{H} \cdot \vec{B} - \frac{1}{2} \oint \mathrm{d}\vec{S} \cdot (\vec{A} \times \vec{H}). \end{aligned} \tag{79}$$

Note that, the second term goes to zero once we look for large volumes[14]. Now the stored magnetic energy becomes

$$U = \frac{1}{2} \int \mathrm{d}^3\vec{r}\ \vec{B} \cdot \vec{H}, \tag{80}$$

the $\frac{1}{2}\vec{B} \cdot \vec{H}$ is naturally the energy density of magnetic field.

6 Maxwell's Equations and Waves

Equipped with all the knowledge of electrostatic and magnetostatic fields, we are finally able to generalise the Maxwell's equations, in both integral and differential forms.

$$\begin{aligned} &\nabla \cdot \vec{D} = \rho_\mathrm{f} && \oint_{\partial V} \vec{D} \cdot \mathrm{d}\vec{S} = \int_V \rho_\mathrm{f}\ \mathrm{d}V \\ &\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} && \oint_{\partial S} \vec{E} \cdot \mathrm{d}\vec{l} = \int_S \frac{\partial \vec{B}}{\partial t} \cdot \mathrm{d}\vec{S} \\ &\nabla \cdot \vec{B} = 0 && \oint_{\partial V} \vec{B} \cdot \mathrm{d}\vec{S} = 0 \\ &\nabla \times \vec{H} = \vec{J} + \frac{\partial \vec{D}}{\partial t} && \oint_{\partial S} \vec{H} \cdot \mathrm{d}\vec{l} = \int_S \left(\vec{J} + \frac{\partial \vec{D}}{\partial t} \right) \cdot \mathrm{d}\vec{S} \end{aligned} \tag{81}$$

[14]In most cases the volume to be integrated is the whole space. $S \propto r^2$, $H \propto r^{-2}$, but $A \propto r^{-1}$, therefore we have at least an r^{-1} proportionality, which converges pretty quickly.

In free space, the equations reduce to

$$\begin{aligned}\nabla\cdot\vec{E}&=0\\ \nabla\times\vec{E}&=-\mu_0\frac{\partial\vec{H}}{\partial t}\\ \nabla\cdot\vec{H}&=0\\ \nabla\times\vec{H}&=\epsilon_0\frac{\partial\vec{E}}{\partial t}\end{aligned}\tag{82}$$

Here is a trick widely used in physics: take the curl of the second equation, we get

$$\nabla^2\vec{E}=-\nabla\times(\nabla\times\vec{E})=\mu_0\frac{\partial}{\partial t}(\nabla\times\vec{H})=\mu_0\epsilon_0\frac{\partial^2\vec{E}}{\partial t^2},\tag{83}$$

which is a wave equation in free space. Similarly, we have

$$\nabla^2\vec{H}=\mu_0\epsilon_0\frac{\partial^2\vec{H}}{\partial t^2}.\tag{84}$$

This is the proof to the famous statement: Maxwell equations indicates that electromagnetic waves may exist, and its wave speed is $c=1/\sqrt{\mu_0\epsilon_0}$ in free space (vacuum).

It can be shown[15] that solution of wave equation is a linear superposition of single-frequency plane waves. Therefore we are only interested in solutions of form $\psi=\psi_0\mathrm{e}^{\mathrm{i}(\vec{k}\cdot\vec{x}-\omega t)}$ unless frequency difference matters.

6.1 Plane wave in isotropic media

From Maxwell's equation, it is clearly seen that EM waves are transverse, and exhibits polarisation. The sign convention from Lenz's law and our choice of phase[16] implies that $\hat{k}=\hat{E}\times\hat{H}$. In this case, it is straightforward to show that

$$\begin{aligned}\nabla\cdot\vec{E}&=\mathrm{i}\vec{k}\cdot\vec{E};\\ \nabla\times\vec{E}&=\mathrm{i}\vec{k}\times\vec{E},\end{aligned}\tag{85}$$

and Maxwell's equations in free space turns into

$$\begin{aligned}\vec{k}\cdot\vec{E}&=0\\ \vec{k}\cdot\vec{H}&=0\\ \vec{k}\times\vec{E}&=\omega\mu_0\vec{H}\\ \vec{k}\times\vec{H}&=-\omega\epsilon_0\vec{E}.\end{aligned}\tag{86}$$

In the handout, the above equations are for amplitudes ($\vec{E}_0$ and $\vec{H}_0$) rather than actual fields, but there's no need to do that, since E and H are in phase.

It is conventional[17] to consider E as the "general force" and H as "general velocity", and it gives the impedance of free space:

$$Z_0=\frac{E}{H}=\sqrt{\frac{\mu_0}{\epsilon_0}}.\tag{87}$$

Note that, the impedance of space does not dissipate power, an EM wave can either supply or carry power away.

In a linear, isotropic and homogeneous non-conducting media, the only adjustment is adding ϵ and μ before every ϵ_0 and μ_0.

[15] Prove by authority.

[16] Phase is written as $\vec{k}\cdot\vec{x}-\omega t$.

[17] Think about their dimensions!

Of course, we can express a general wave in frequency space using Fourier transform,

$$\vec{E}(\vec{x},t) = \int \mathrm{d}^3\vec{k}\ \mathrm{d}\omega\ \vec{A}(\vec{k},\omega)\mathrm{e}^{\mathrm{i}\vec{k}\cdot\vec{x}}\mathrm{e}^{-\mathrm{i}\omega t}, \tag{88}$$

where $\vec{A}$ is called the spectral function. The plane-wave decomposition is the starting point of quantum optics. Each plane wave can be regarded as a simple harmonic oscillator, which must be quantised. Also, it is central to the whole notion of photons in modes, and is used in the derivation of Planck's law for the black body radiation.

6.2 Energy flow

To calculate the total flow of energy, we start with the vector identity

$$\nabla\cdot(\vec{E}\times\vec{H}) = \vec{H}\cdot(\nabla\times\vec{E}) - \vec{E}\cdot(\nabla\times\vec{H}),$$

and apply Maxwell's equations to get

$$\nabla\cdot(\vec{E}\times\vec{H}) = -\mu_0\vec{H}\cdot\frac{\partial\vec{H}}{\partial t} - \epsilon_0\vec{E}\cdot\frac{\partial\vec{E}}{\partial t} - \vec{E}\cdot\vec{J}. \tag{89}$$

Taking the volume integral, one should realise that the result is just the change in total energy of the field inside that volume.

$$-\oint_{\partial V}\mathrm{d}\vec{S}\cdot(\vec{E}\times\vec{H}) = \int_V \mathrm{d}^3\vec{r}\ \left[\frac{\partial}{\partial t}\left(\frac{1}{2}\mu_0\vec{H}\cdot\vec{H}\right) + \frac{\partial}{\partial t}\left(\frac{1}{2}\epsilon_0\vec{E}\cdot\vec{E}\right) + \vec{E}\cdot\vec{J}\right]. \tag{90}$$

Therefore, $\vec{E}\times\vec{H}$ should be some kind of energy flux flowing into the volume, which is called the Poynting vector,

$$\vec{N} = \vec{E}\times\vec{H}. \tag{91}$$

6.3 Radiation pressure and momentum

From special relativity, E is simply pc for photons since they have no rest mass. The radiation energy density U is defined by the radiation momentum density $\vec{g}$ times the light speed c,

$$U = |\vec{g}|c. \tag{92}$$

Using the Poynting vector, $\vec{g}$ can be expressed as $\vec{N}/c^2$. Since the radiation pressure is defined by the rate of change of momentum per unit area, it can be expressed as

$$\vec{R} = \frac{\vec{N}}{c}. \tag{93}$$

Note that, we have assumed that radiation passes through the surface. If the surface reflects the radiation instead, the pressure should be doubled.

6.4 Complex power

As explained in Physics A course, the complex power must be defined by

$$P = \frac{1}{2}V^*I. \tag{94}$$

Another way to think about this is to consider our new definition of dot product, which is discussed in the mathematics course.

The real part (real) of the complex power gives the average rate at which energy flows *into* the component. The imaginary part (reactive) gives the maximum rate at which energy sloshes in and out of the component. Also, power flow is distinct from the complex representation of the current and voltage, if two systems are connected together, what we need to do is always simply add the complex powers.

6.5 Reflection and transmission

Snell's law, together with the reflection and transmission coefficients, can be derived from the boundary conditions. We shall not repeat the derivation here.

The only thing to note is, at a certain angle called Brewster angle,

$$\tan\theta_{\mathrm{B}} = \frac{n_2}{n_1}, \tag{95}$$

the parallel reflection coefficient can be zero. This angle are often used to eliminate reflections from optical windows or to produce linearly polarised light from unpolarised light.

And, don't forget, we still have the critical angle $\theta_{\mathrm{C}} = \arcsin n_2/n_1$, which eliminates the transmission wave when θ_i exceeds this angle. In electrodynamics, we will see that an evanescent wave is produced that decays exponentially from the surface.

6.6 Waves in plasmas

Up to this point, we have considered waves in some non-conducting medias. Now, we shall consider waves in a special kind of conducting media: plasmas.

Plasmas are materials that have perfectly free electrons and their parent ions (to make it neutral overall). Since the ions are heavy, we shall simply ignore their motions.

The equation of motion for an electron is given by the Lorentz force

$$m_{\mathrm{e}}\frac{\mathrm{d}^2\vec{r}}{\mathrm{d}t} = -e(\vec{E} + \vec{v}\times\vec{B}). \tag{96}$$

But we assume that v is a lot smaller than the speed of light, and $\vec{E}$ has the form of a plane wave

$$\vec{E} = \vec{E}_0\exp[\mathrm{i}(kz-\omega t)].$$

Therefore we can ignore the second term at the RHS of the equation of motion, and simply write

$$\vec{r} = \frac{e}{m_{\mathrm{e}}\omega^2}\vec{E}_0\exp[\mathrm{i}(kz-\omega t)]. \tag{97}$$

Suppose the plasma at rest is everywhere locally neutral (which is impossible in real materials), the dipole moment density induced by the electric field is simply

$$\vec{P} = -Ne\vec{r} = -\frac{Ne^2}{m_{\mathrm{e}}\omega^2}\vec{E}, \tag{98}$$

where N is the number of electrons per unit volume. Recall the relation $\vec{P} = \epsilon_0(\epsilon-1)\vec{E}$, the relative permittivity of plasma is therefore

$$\epsilon = 1 - \frac{\omega_{\mathrm{p}}^2}{\omega^2}, \tag{99}$$

where the plasma frequency ω_{p} is defined as

$$\omega_{\mathrm{p}}^2 = \frac{Ne^2}{\epsilon_0 m_{\mathrm{e}}}. \tag{100}$$

At frequencies below the plasma frequency, EM waves are reflected off, and cannot transmit through the plasma. That somehow benefits technology because EM waves can reflect off the ionosphere, enabling long-range communications.

The refractive index becomes a function of ω_{p},

$$n = \sqrt{\epsilon} = \left(1 - \frac{\omega_{\mathrm{p}}^2}{\omega^2}\right)^{1/2}. \tag{101}$$

If n is imaginary, $k = n\omega/c$ is of course imaginary, leading to the exponential decay we expected:

$$\vec{E} = \vec{E}_0 \exp\left[-nk_0 z\right] \exp\left[\mathrm{i}\omega t\right]. \tag{102}$$

The lower ω is, the shorter the distance over which the field decays. Note that, the decaying wave is evanescent (not propagating).

If we check the magnetic field and calculate the complex power, we will find that $\vec{E}$ and $\vec{H}$ are out of phase, meaning that there is no net transfer of energy within the plasma, and energy sloshes in the z direction at any point. This implies that energy is stored in the motion of the oscillating electrons, and all of the energy in the original wave must be reflected by the plasma (on average).

Above the plasma frequency, we can calculate the dispersion relation because k is already expressed as a function of ω. It is easy to find that $v_{\mathrm{p}} v_{\mathrm{g}} = c^2$.

6.7 Waves in conducting media

In an ideal metal, density of electrons is high, so the scattering effect becomes significant, therefore we are forced to consider Ohm's law, $\vec{J} = \sigma\vec{E}$. Using Maxwell's equations,

$$\nabla \times \vec{H} = (\sigma - \mathrm{i}\omega\epsilon\epsilon_0)\vec{E} = -\mathrm{i}\omega\epsilon_0 \left(\epsilon + \frac{\mathrm{i}\sigma}{\omega\epsilon_0}\right)\vec{E}. \tag{103}$$

The effective dielectric constant is therefore related to conductivity by

$$\epsilon' = \epsilon + \frac{\mathrm{i}\sigma}{\omega\epsilon_0}. \tag{104}$$

A material behaves as a dielectric if the real part of the complex dielectric constant dominates, while it behaves as a metal if the imaginary part dominates (high conductivity). Therefore, in practical cases we often ignore the real part if we are focusing on a well-behaved metal.

For a good metal, the refractive index is

$$n = \sqrt{\epsilon'\mu} = \pm\frac{1+\mathrm{i}}{\sqrt{2}}\sqrt{\frac{\sigma\mu}{\omega\epsilon_0}}. \tag{105}$$

Again, plug this into $k = n\omega/c$, we have

$$k = \frac{1+\mathrm{i}}{\delta}, \quad \delta = \sqrt{\frac{2}{\sigma\omega\mu_0\mu}}, \tag{106}$$

and δ is called the skin depth because it is the decay constant of the wave

$$\vec{E} = \vec{E}_0 \exp\left[-\frac{z}{\delta}\right] \exp\left[\mathrm{i}\left(\frac{z}{\delta} - \omega t\right)\right]. \tag{107}$$

We've taken the plus sign in the expression of n because the other solution (exponentially increasing) is not physical.

Again, we shall consider the energy flux in a metal. For a perfect metal $\vec{E}$ and $\vec{H}$ are 45° out of phase, therefore Poynting vector can take positive and negative values, depending on time. On average, the power still travels forwards, and dissipates ohmically into heat inside the conductor.

6.8 The skin effect

The skin effect only takes place in AC circuits. Consider a wire carrying current I that oscillates at frequency ω. The electric field is always aligned with the current (along the wire). The magnetic field generated will induce another electric field that opposes the original field, this effect is maximised at the centre (Lenz's law). Effectively, the current will all flow at the skin of the wire. Considering the Poynting vector, the energy flows into and out of the surface of the wire.

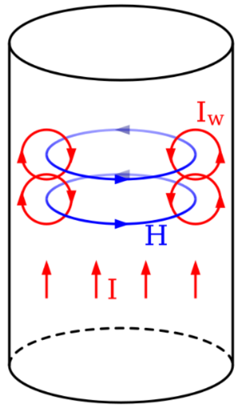

Figure 1: Skin effect, the direction of current (E field), H field, and the induced current I_w are shown.

If z is pointing to the axis of the wire from its surface, J can be expressed using $J = \sigma E$,

$$J_z = J_0 \exp\left[-\frac{z}{\delta}\right] \exp\left[\mathrm{i}\left(\frac{z}{\delta} - \omega t\right)\right]. \tag{108}$$

As expected, the skin depth decreases as the frequency increases, therefore the resistance increases.

If $\delta \ll a$, a is the radius of the wire, a good approximation when calculating the total current is to integrate z from zero to infinity. This makes everything easier because exponentials are always easy to integrate.

At very high frequencies, electrons move only for a short distance per cycle, which means scattering effects are greatly reduced. Then metals behave just like plasmas.

Another special case is the superconductors. They have a particularly high kinetic inductance term: undamped motion of Cooper pairs. The behaviour of superconducting transmission lines is modified significantly by this kinetic inductance effect.

7 Guided Waves

Due to the skin effect, wires become lossy at high frequencies. If the circuit dimension d is comparable to or larger than a given wavelength, the assumption that the voltage and current are constant along the wire breaks down. Luckily, transmission lines ($\lambda \simeq d$) and waveguides ($\lambda \ll d$) can still guide the wave and work as "wires".

For example in a microwave (several GHz), we must use waveguides, while in CPUs ($\sim$ 1 GHz), the dimension is small compare to wavelength, therefore only classical wires are required.

7.1 Transmission lines

First we consider a transmission line made up of a pair of wires, z-axis is along the wire, and we have voltage across the wires, $V(z)$, and current $I(z)$ along the wire.

Assume that the wire has inductance L and capacitance C per unit length. The transmission lines are modelled as a series of inductors and a parallel row of capacitors. The voltage drop and current change are therefore

$$\begin{aligned} \frac{\partial V}{\partial z} &= -L\frac{\partial I}{\partial t}; \\ \frac{\partial I}{\partial z} &= -C\frac{\partial V}{\partial t}. \end{aligned} \tag{109}$$

These equations are valid for any transmission line that supports a transverse electromagnetic wave (V and

I in phase). From these equations it is also clear that

$$\begin{aligned}\frac{\partial^2 V}{\partial z^2} &= LC\frac{\partial^2 V}{\partial t^2};\\ \frac{\partial^2 I}{\partial z^2} &= LC\frac{\partial^2 I}{\partial t^2}.\end{aligned} \tag{110}$$

The current waves that travel along the line has velocity $v = \pm 1/\sqrt{LC}$. The impedance is $V/I = \sqrt{L/C}$.

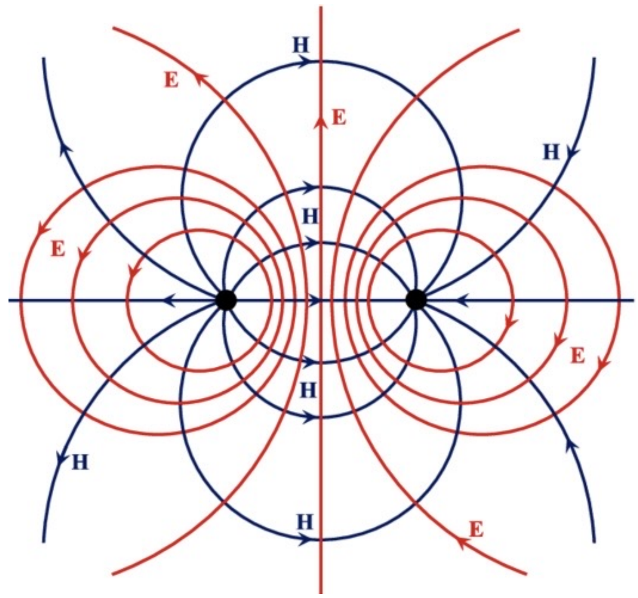

Figure 2: Directions of H and E fields generated by parallel wires, showing that the Poynting vector always points along the wire. The wires are embedded in dielectric materials, where EM wave propagates.

Consider a semi-infinite transmission line connected to a power source, the effect is the same as a resistor of its characteristic impedance. In engineering cases, a load is often impedance-matched to the transmission line, to reduce energy loss and reflections.

For a parallel line system, the capacitance and inductance were found to be

$$C = \frac{\pi\epsilon_0}{\ln(2D/a)}; \quad L = \frac{\mu_0}{\pi}\ln(2D/a). \tag{111}$$

Plug into the equation for wave speed, and notice that $v = c/n$ for the surrounding dielectric materials. It is easy to show that

$$Z = \frac{Z_0}{n}\frac{\ln(2D/a)}{\pi}. \tag{112}$$

If the wires are not fully embedded in dielectric materials, i.e. $\sqrt{\epsilon}$ has complex term, the transmission lines can't support a transverse TM wave. However, the wave usually behaves in a transverse-like manner as long as the frequency is low enough. To reduce coupling noise, what we need to do is twist the parallel wires together, as investigated in practical sessions.

For coaxial cylinders, which is widely used in cables such as BNC, the result is similar. But any azimuthal structure in the dielectric material will prevent a pure transverse EM wave from propagating. Normally, commercial coaxial cylinder cables have characteristic impedance 50 Ω.

Strip transmission line is another commonly used example, which links to printed circuit boards (PCBs) running at high frequencies.

By changing d and a in the strip transmission line, we can change its characteristic impedance. In real life, edge effects will appear and the conductors are separated by a dielectric half space rather than completely

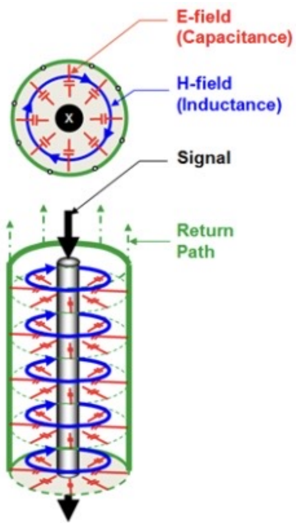

Figure 3: E and H fields of the coaxial cylinders, which supports transverse EM waves.

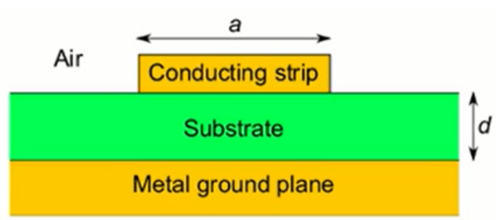

Figure 4: Cross-section of a strip transmission line, normally $d \ll a$, so we can ignore edge effects.

embedded, so the wave insider the lines is not perfectly transverse EM wave. However, the TEM analysis is still good enough if because the frequency is high and the dielectric thickness is small.

Another type of stripline, with less crosstalk effect, is a coplanar waveguide. This is made by simply adding grounded conductors beside the conducting strip. This reduces the edge effects and therefore support TEM better.

7.2 Power flow in transmission lines

We consider the energy stored in the potential difference (rather than in the field, which is hard to calculate). The rate of change of stored energy is simply

$$\frac{\mathrm{d}U}{\mathrm{d}t} = \int_a^b \mathrm{d}z \, \left(LI\frac{\partial I}{\partial t} + CV\frac{\partial V}{\partial t} \right). \tag{113}$$

From the wave equations, the time integrals can be changed into length integrals,

$$\frac{\mathrm{d}U}{\mathrm{d}t} = -\int_a^b \mathrm{d}z \, \left(I\frac{\partial V}{\partial z} + V\frac{\partial I}{\partial z} \right). \tag{114}$$

Therefore the integral evaluates to

$$\frac{\mathrm{d}U}{\mathrm{d}t} = [VI]_a - [VI]_b. \tag{115}$$

The rate of change of the stored energy is equal to the rate at which energy flows in at point a minus the rate at which energy flows out at b.

Consider a normal setup, the transmission line carries energy to a load R. If impedance is matched, $R = Z$, no power is reflected, therefore all energy flows into the load. But reflection will occur if they are not matched.

Suppose V_t is the voltage across the load, I_t the current flowing through it, and Z_t its impedance. We will carry on the notation Z for characteristic impedance of the transmission line. Suppose we have incident wave V_i, I_i and reflected wave V_r, I_r, they must be in-phase and have opposite sign on k. The waves are linked by the continuity of voltage and current, and further connected by the relation $V_t/I_t = Z_t$.

The solution is straightforward, voltage reflection and transmission coefficient are

$$r = \frac{Z_t - Z}{Z_t + Z}; \quad t = \frac{2Z_t}{Z_t + Z}. \tag{116}$$

7.3 Input impedance of transmission lines

We assume that the length of the line is a, and Z_{in} is the input impedance (impedance of the power source). The point between source and the line is at $z = -a$ compare to the reference point (point between the line and the load). Therefore

$$Z_{\text{in}} = \frac{V_i + V_r}{I_i + I_r} = \frac{\mathrm{e}^{\mathrm{i}ka} + r\mathrm{e}^{-\mathrm{i}ka}}{\mathrm{e}^{\mathrm{i}ka} - r\mathrm{e}^{-\mathrm{i}ka}} Z, \tag{117}$$

plugging in the value of r, calculated above, we have

$$\frac{Z_{\text{in}}}{Z} = \frac{Z_t \cos ka + \mathrm{i}Z \sin ka}{Z \cos ka + \mathrm{i}Z_t \sin ka}. \tag{118}$$

Memorising this expression is highly recommended.

Consider some special cases:

- If the circuit is shortened, Z_{in} is purely imaginary, all of the power is reflected.
- If the circuit is open-looped, Z_{in} is also imaginary, all of the power is reflected.
- If $a = \lambda/4$ (quarter-wavelength line), $Z_{\text{in}} = Z^2/Z_t \in \mathbb{R}$, there are no reflections. Therefore, in a transmission line system, impedance matching will only be effective at one particular frequency (correspond to the distance).

Whether the circuit is inductive or capacitive depends on the sign of imaginary part. And multiple lengths of transmission line having different characteristic impedances can be connected together to achieve broadband matching[18].

7.4 Waveguides

Transmission lines support a transverse EM wave, or at least a wave that is very close to being TEM for the frequency of operation, but it will give a high loss at high frequencies. Waveguides solve this problem by not requiring TEM, and not requiring a second conductor at the same time. Therefore we move from considering V and I to $\vec{E}$ and $\vec{H}$, propagating as (approximate) plane waves in wave guides.

It is not hard to visualise that two plane waves of the same frequency and 2θ between the wave numbers will add, and create a standing component and a moving component. Plus, the two components are perpendicular to each other.

In the moving direction, phase velocity would give $v = c/\cos\theta > c$ but it doesn't violate special relativity. The group velocity is $v_{\mathrm{g}} = c\cos\theta$ which is smaller than the speed of light.

We want to place conductors along the moving direction, so that the conductors lie on the positions where the standing part gives zero. In this way, plane waves at a certain angle θ can propagate along the waveguide.

From the group velocity, the group wave vector is $k_{\mathrm{g}} = k\cos\theta$, and its relation with the standing part wave number $k\sin\theta = n\pi/a$ is

$$k_{\mathrm{g}}^2 = k^2 - \frac{n^2\pi^2}{a^2}. \tag{119}$$

[18] Consider the coating on camera lenses and galsses.

Of course, in order to set up a capable waveguide, we must have

$$m\lambda/2a < 1. \tag{120}$$

We refer to the modes using notations such as TE_{10}, this means $n = 1$ in the standing wave direction, and no variation in the remaining direction ($m = 0$). TE stands for transverse electric because E field only points to the remaining y direction.

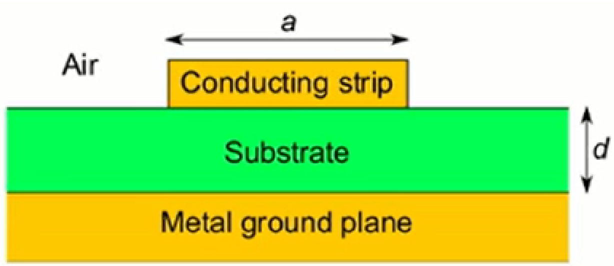

Figure 5: $\vec{E}$ (dotted) and $\vec{H}$ (solid) fields of the TE_{10} mode.

However, H field is not transverse since it points to both the x and z directions. It is not zero at the surface of the top and bottom conductors, and gives rise to a surface current (and heat of course).

7.5 General TE modes

We can allow $\vec{E}$ to vary with both x and y across the waveguide, whilst remaining transverse electric wave. The general solution is

$$\begin{aligned} E_x &= A_0 k_y \cos k_x x \sin k_y y \cos(k_z z - \omega t) \\ E_y &= -A_0 k_x \sin k_x x \cos k_y y \cos(k_z z - \omega t) \\ E_z &= 0 \end{aligned} \tag{121}$$

where

$$(k_x, k_y, k_z) = \left(\frac{n\pi}{a}, \frac{m\pi}{b}, k_{\mathrm{g}}\right). \tag{122}$$

This mode is called TE_{nm} mode, where we require m and n can't both be zero. To satisfy the wave equation, we must have

$$k_z^2 = k_{\mathrm{g}}^2 = \frac{\omega^2}{c^2} - \frac{m^2\pi^2}{a^2} - \frac{n^2\pi^2}{b^2}. \tag{123}$$

And if we define the critical wave number $k_c^2 = k_x^2 + k_y^2$, which is fixed using a pair of $(n.m)$, we arrive at the waveguide equation

$$k_{\mathrm{g}}^2 = |\vec{k}|^2 - k_c^2. \tag{124}$$

The critical wave number also provides us with the cut-off frequency $\nu_{\min} = ck_c/2\pi$. Below this frequency, wave cannot propagate at this given mode.

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